The Lies of Aggregate Indicators

It is impossible to open a newspaper or article in sections relating to current economic or financial trends without being bombarded by numbers and complex-sounding jargon, usually accompanied by some indicator not well-understood by the average reader. These indicators are often abbreviated by what looks like alphabet soup – GDP, GNP, GDP per capita, AGR, CPI, PMI, and heaven knows what else[1]. Even when the indicators are written in natural language terms – interest rates, stock market prices, inflation rates, government spending, unemployment, budget deficits, trade surpluses – it is still unclear to the uninformed reader what these terms mean. Even if a reader is informed about the topic, he or she may not fully understand the real-world implications from the numbers being presented to them. What does it mean on an individual level if "inflation is rising", thereby "decreasing the average purchasing power of disposable income", when at the same time "GDP is rising" and "the economy is growing at 4.5% per annum"?

This project aims to look at the very common GDP, used as a measure of national wealth, and contrast it to actual incomes of the individuals making up the nation. I became interested in this discussion after learning how economic growth rates oftentimes do not reflect deeply polarised income inequalities in a nation[2]. Specifically, corporate and billionaire wealth exceeds that of the ordinary individual by so far that on average, even if the majority are barely surviving or exist below the conditions necessary for continued, healthy lives, the country's overall assets may simultaneously show increases.

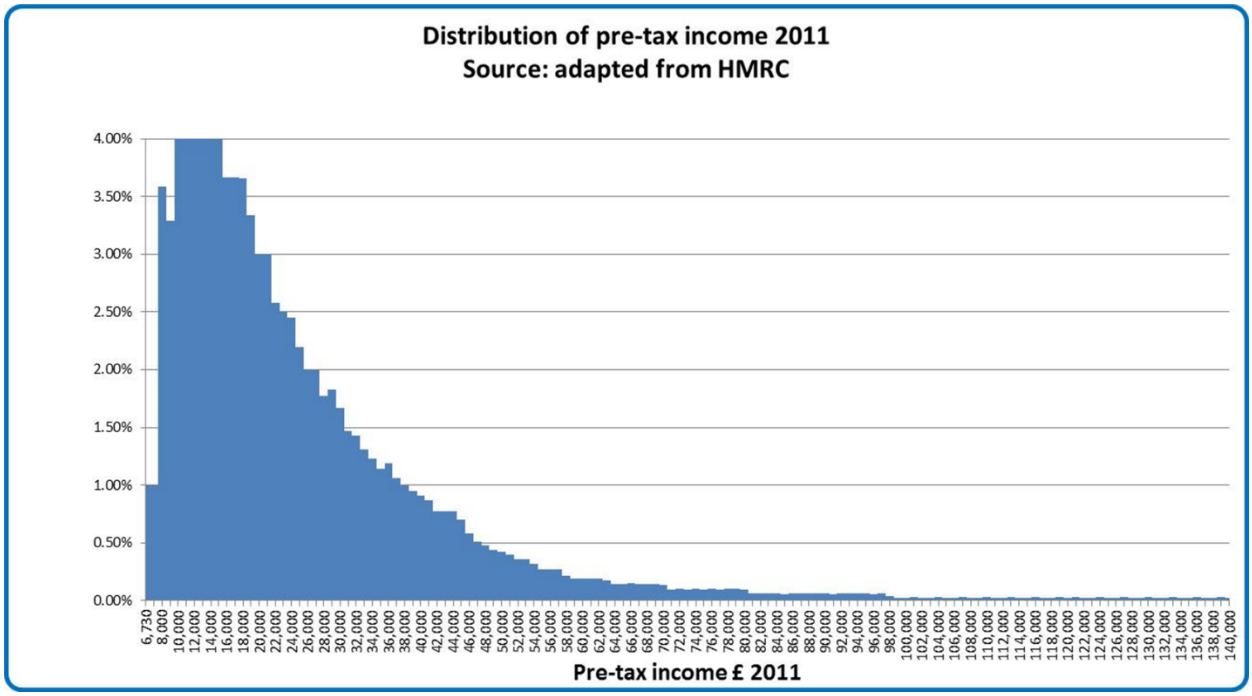


Fig. 1*: Pre-tax income in the UK, 2011*

For instance, in the above histogram, it is painfully clear that the vast majority of people earned below 50,000 GBP in 2011. However, despite the miniscule fraction of the

population that earned above 100,000 GBP, if an arithmetic mean of all incomes is calculated, the discrepancy between the income class with the largest number of members and the skewed tail-end is high enough to result in a very biased average. In this instance, the mean was reported at almost 60,000 GBP, while the median is actually under 20,000 GBP – as can be seen from the graph, 60k is far beyond the income range of most individuals, with the modal incomes by far surrounding 15,000 GBP. Adding outlying values – very large or very small values that reflect less than a small fraction of the population – to the overall sum when calculating the arithmetic mean can inflate the final answer significantly. For this reason, most modern economic analyses use median household income rather the mean, since the median is only an indicator of the central neighborhood (assuming there is one) around which the largest frequency of data values may be found. Yet somehow we have not made a similar transition for GDP – GDP is still calculated as an aggregate of all production in the economy, and as such may not accurately reflect the state of people across various socio-economic strata. In comparing GDP to Median Income, we see how an overall measure of economic affluence compares to actual income levels and standards of living of people living their everyday lives.

The CPI, Consumer Price Index, is meant to track the average change over time of a standard basket of goods in the market. It is primarily affected by market changes and over time, inflation. Any meaningful discussion of income must contrast income to CPI to get an understanding of how much someone's income is actually worth – how much can someone actually buy with the income that they have? When we hear the term "adjusted for inflation" with regard to some amount of income, that is the standard adjusting for inflation is trying to set: $1 in 1990 may not be able to buy the same amount as $1 in 2020, and the above adjustment equates both to some pre-specified dollar standard. It demonstrates an average individual's "real purchasing power".

These concepts are well demonstrated in this graph below of change over time in the GDP and median household income (both adjusted for inflation) in the US from 1993 onwards. The trend-lines demonstrate that while GDP was rising significantly, indicating larger economic output from American businesses and industries, the median household income by contrast has not reached its pre-2000 peak since and is still actively falling. As Nobel Laureate in Economics and former vice president of the World Bank Joseph Stiglitz argues[4], this kind of discrepancy between GDP and income is indicative of a polarised income spectrum. Large businesses in the hands of a few are growing and are responsible for most of the GDP growth, while the average American is actually not being able to buy as much as they used to be able to. When prices rise and overall income stays the same, the amount one is actually able to buy, one's real income, ends up decreasing. A fast-growing GDP by itself is thus a woefully inadequate indicator of most people's reality.

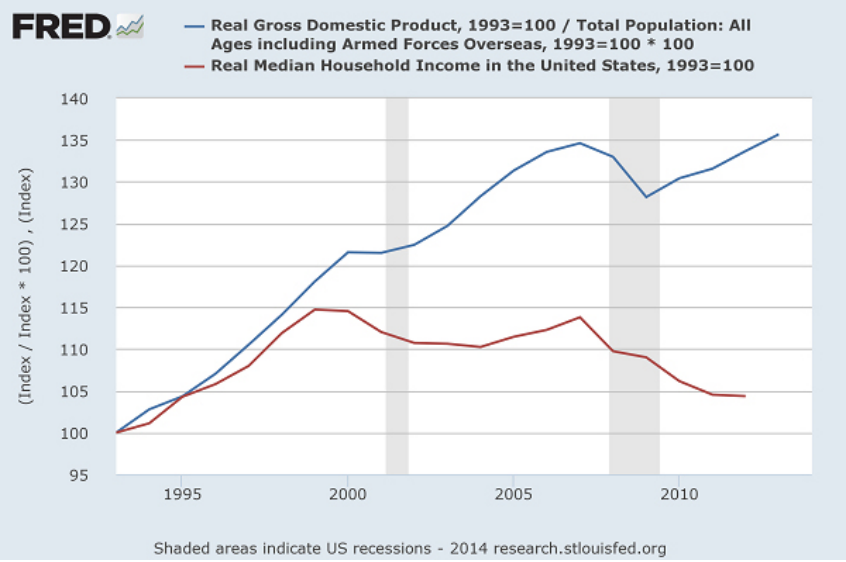


Fig. 2*: Real GDP vs. Real Median Household Income in the US*
Taken from FRED Blog, Economic Research Dept. at Federal Reserve Bank of St. Louis

This shortcoming comes from GDP being commonly calculated in the expenditure approach as GDP = C (all private sector consumption) + G (total government expenditures) + I (sum of investments in capital, inventories and housing) + NX (net exports, or total exports minus total imports). Summation of sales does not track values like sustainable development, minimum standards of living, and malpractices followed by those attempting to maximise some form of profit or GDP-related metric. Unregulated free-market advocates may defend this by saying something like “negative externalities can be tracked and minimised in cases where the economic costs are significant”, but again, as I have stated above, not all values are so easy to quantify. The cost to a community of a factory making a river unsuitable for fish breeding is more than simply the sales price of all the fish that could have been sold; and in more complex cases, how does one even begin to quantify the negative externalities of people dying because their real median incomes are decreasing?

I will attempt to quantify this discrepancy by creating a probability distribution that tracks individual ability to lead a healthy life. I fully recognise that a combination of all the relevant factors that influence such a metric is a comprehensive study far beyond the scope of my current project, so I will simply be looking at a distribution defined by real income. A natural extension of this work would be to construct a similar model but one that considers many more of the other factors that may affect an individual’s ability to achieve the things they want to, such as access to healthcare, education, potential for social mobility, etc. But the creation of such a metric would require the combination, for each individual, of their current state in each of these factors, which is a 10 to 20 factor weighted index for millions of data points. Given my lack of access to individual data and the scale of such a task, I am restricting my focus to aggregated real income data. Ultimately, one’s income levels directly affect the opportunities and resources one has access to in a capitalistic

economy anyway, so a single factor analysis should still be good enough to demonstrate the discrepancies I want to show.

There is a commonly understood mathematical definition of a continuous probability function f(x):

P(a < x < b), the probability that x lies within a certain set of bounds, is given by $\int_a^b f(x)\,dx$

where f(x) is non-negative for all real x, and $\int_{-\infty}^{\infty} f(x)\,dx = 1$; i.e. all possible outcomes from the probability function sum together to provide a probability of 1. The total probability of x being some value is 1: x must be some value between negative infinity and positive infinity.

The value of p(x = a) for any specific value of a will always be practically 0. A probability function is not a typical function in that f(x) will give you an output value for some input value of x (that's discrete thinking); rather, the way to commonly understand the values from a probability function is that f(x) for some specific value v is the probability that our variable x lies between some reference point on the x-axis, say 0, and the value v. Probability function f(x) is thus a measure of where we are most likely to find our values of x. Below is an instance of what a symmetrical standard normal curve looks like:

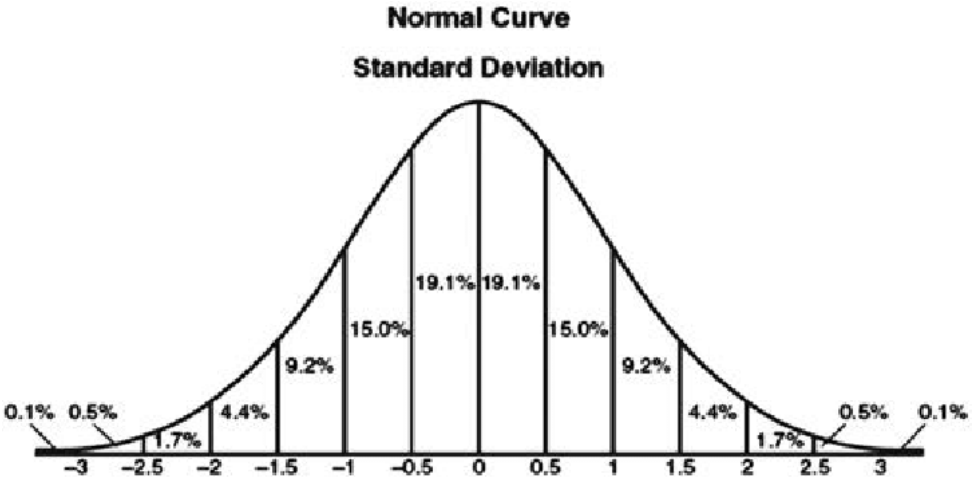


Fig. 3*: A standard normal probability distribution*
Photo from researchgate.net

Given the complexity of this discussion and the massive amount of work involved in creating a comprehensive, continuous, integrable probability distribution, I have instead opted for my purposes to use a frequency table, from which will be derived a histogram that represents the underlying probability distribution. In order to revert from a discrete, frequency-based histogram built upon continuous data to an actually continuous probability model of the underlying data, once I have the histogram, I will plot specific points from it on a spreadsheet and attempt several regressions. Naturally, the models can be re-expressed with necessary scale factors afterwards to ensure total area under the graph remains equal to 1; as such, my focus will be more on the shape of the

distribution than on the exact equation (the exact equation is unlikely to be too accurate anyway, given that it was not built with access to each individual data point).

Nonetheless, I believe the histogram will still serve to make the point I wish to illustrate: if f(x) represents the probability distribution that a person's economic strength, x, will be found within a certain range, most people will be clustered around relatively low values of x. Although I further suspect that the average person's civil and political strength – defined, for instance, as the likelihood of their being able to influence policy decisions that affect them, their family and their community – will be even lower and almost uniformly zero for the vast majority, given that I do not currently have the means to quantify or the data to back up such a suspicion, I will content myself with tracking income and demonstrating the inadequacies of using a single measure of central tendency to summarise variance across a multi-modal distribution.

Let us start this analysis by importing a data table (see appendix) from the US Census Bureau that reflects the percentage of households in the US that fall within a certain income range.

A histogram from this data for the year of 1967 follows here:

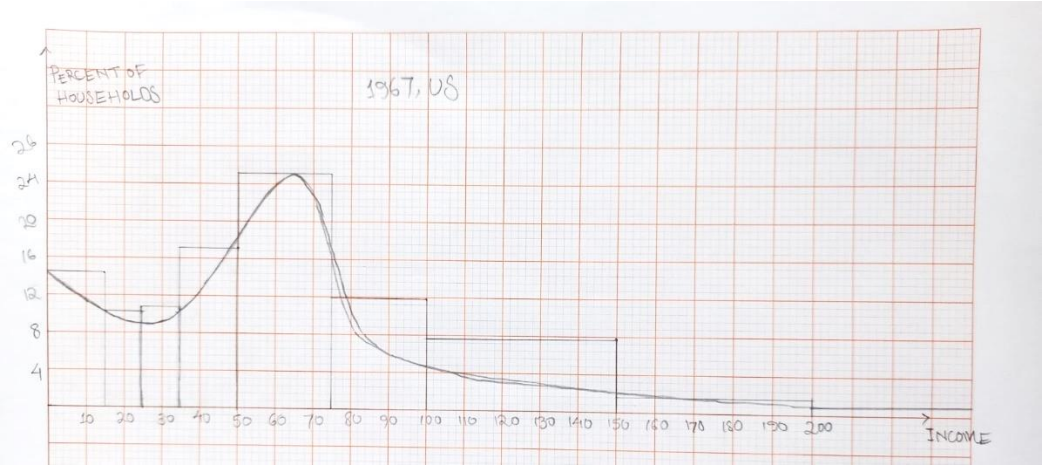


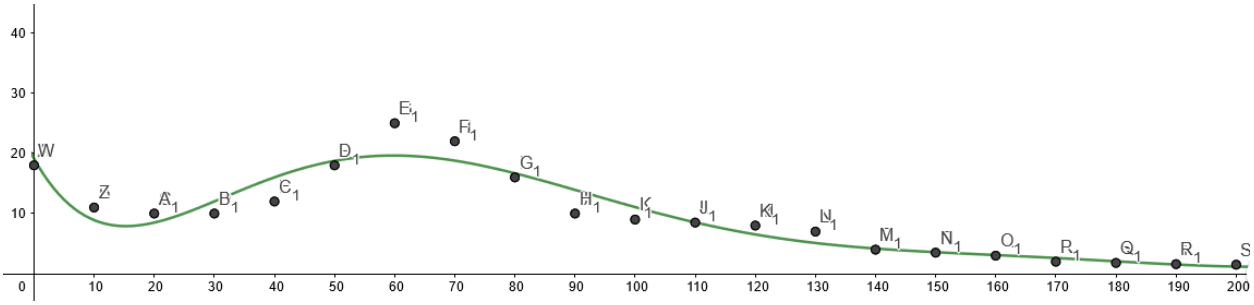
Fig. 4*: Percent of Households at Specified Income Levels, 1967*

The x-axis for these histograms contains income values in thousands of dollars, while the y-axis represents the percentage of households that fall into a certain income bracket. The curve is the estimated continuous probability distribution based on the discrete data. It is an estimate because I did not have access to the individual data points, so the exact shape of the curve is a best fit guess from aggregates.

This graph is highly skewed to the right. From about the 65k range onwards, it can almost be thought of as a normal or t-distribution with low standard deviation. The sharp spike in households below 15k makes the distribution look like it should really start somewhere around the 25k range but had the section before 15k just added on unexpectedly. This is the portion of people living in absolute poverty (off incomes too low to support healthy lives with basic survival needs met), and it is far higher than the section of people who

are living on more than 80k USD per annum. Contrasting with CPI data from that time reveals that this means there were more people below the poverty line than there were in a solid middle class (defined as being able to afford education at a good college and the costs of healthcare), with the large majority of people living paycheck-to-paycheck and just about making ends meet with 50k per annum. It is worth remembering that this data is for households and not individuals, so these earnings are contributing to the expenditures of, in general, between 3 to 4 people.

Putting the estimated curve of best fit through analysis software to arrive at a single function that fits this curve, however, did not seem like a good idea because of the appearance of multiple modes in each graph. Multi-modality is generally an indication of sections of the population that are in different circumstances and should be understood via sample-specific metrics, probably in separate calculations altogether, not least because modelling multiple maxima and turning points precisely with polynomial functions is nowhere near easy. Still, attempting a standard polynomial regression from the outset yields:



with the following equation determined by GeoGebra:

$$-0.000000000000239\,x^{7.00} + 0.000000000229\,x^{6.00} - 0.0000000886\,x^{5.00} + 0.0000175\,x^{4.00} - 0.00182\,x^{3.00} + 0.0926\,x^{2.00} - 1.78\,x + 19.1$$

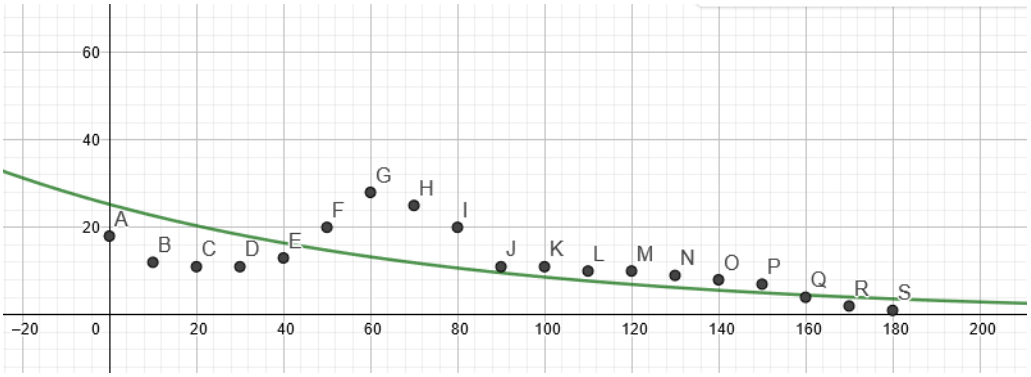
I did not use a higher degree polynomial because doing so resulted in large spikes and falls towards the higher end of the distribution. While the area under this particular curve is closer to 100 than 1, this is easily adjusted by multiplying the entire function by a factor of 1/100 to convert from a percentage to a proportion. Furthermore, the r^2 value of this graph is fairly high including the tail, but if it is only calculated for the graph prior to income levels of 100k, the r^2 value becomes significantly smaller, indicating the polynomial does not entirely accurately reflect the distribution of incomes near the modal bracket, as the error table in the appendixed Excel file demonstrates (it uses a power six polynomial, with similar results). This means adopting polynomial models of high powers risks alienating our ability to understand income distributions near the very low incomes.

Furthermore, this model has seven coefficients, which would be very difficult, if not impossible, to conceptually relate back to real-world factors that determine income levels, not least because each of those factors would have to be related to income proportionally to their square, their cube, etc., all the way up to the power 7. In fact, this polynomial looks more like a Taylor series transform of some transcendental function, in which case we

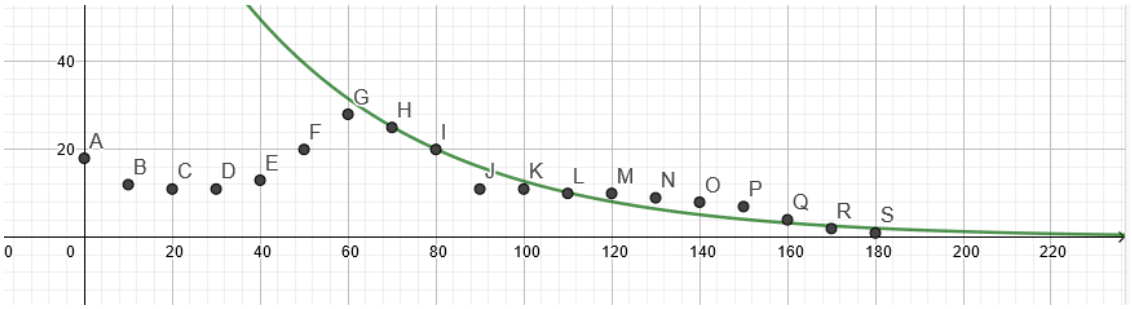
are probably better off ignoring the many-degree polynomial entirely and setting out in search of the underlying transcendental function.

It is further worth noting here that a polynomial curve can only ever approximate a probability distribution, as a distribution that is asymptotic must have exponential, rational, or logarithmic components. In this case, so long as less than 1% of the total is represented in the area beyond 250k on the income range (for which we have insufficient data to predict trends and specifically point out billionaires, as they are all grouped into the category of beyond 250k), for the majority of the distribution the polynomial may still be a good approximation. So to exponentials we now turn, but initially with somewhat disappointing results, for reasons further elucidated below.

Before I attempt to fit another model onto this data, I should reiterate here my strong doubts about the fruitfulness of what I am currently attempting. The initial graph makes it pretty clear that there are two separate modes, and as such, I feel crafting a piecewise function with a cutoff point somewhere halfway between the two modes would yield significantly better results. However, as my project is all about demonstrating the discrepancies that result from lumpsum analyses, I should still like to demonstrate what a variety of non-piecewise regressions for this distribution might look like, the better to pinpoint exactly what about these real-world situations such models miss.

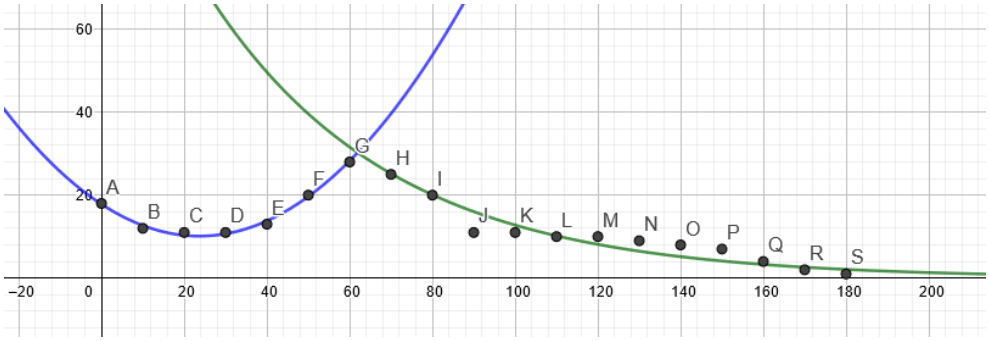


As is clearly visible, the exponential f(x) = $25.27\ e^{-0.01x}$ models the data fairly poorly (the r^2 value is ridiculously low, and understandably so, given that the exponential completely fails to predict the existence of the entire modal middle-income classes). But the exponential does seem to do a relatively good job of modelling the skewed running tail, so I wondered what the regression might look like if I only attempted to apply this exponential past the local maximum at 60k:



From 60k onwards, $122.71\ e^{-0.02x}$ is a much better model with an acceptably high r^2 value. But this still leaves the points upto 60k unmodelled.

Given the local minimum, I thought I might start with a quadratic:

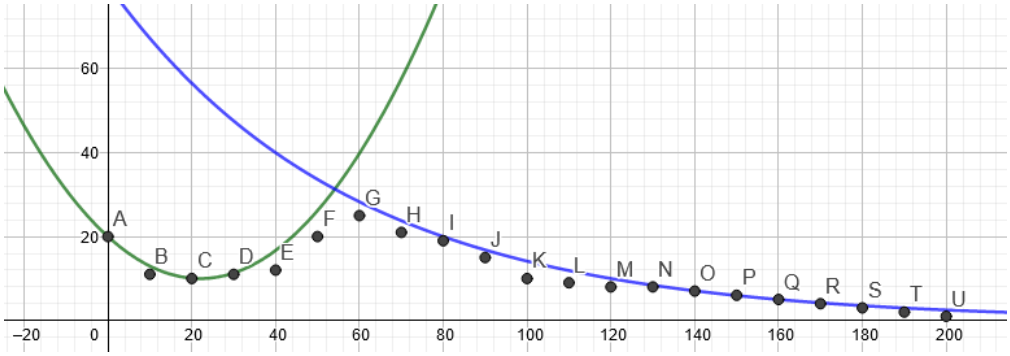


This, in comparison to the initial ugly uncorrelatable 7th degree polynomial model, is absolutely gorgeous, with mean square of errors at only 1.5% for any given data point (see appendixed Excel). The quadratic f(x) = $0.01\ x^2 - 0.65\ x + 17.82$ has an extremely high r^2 value for income brackets up to 60k; a little smoothing and data re-expressing can yield even better results or a way to meld the two definitions for this piecewise function into a single cohesive whole. At the least, this is conceptually significantly better than the 7th degree polynomial: for instance, the quadratic may be expressed as $a(x - b)^2 + c$, where (b, c) being the turning point of the polynomial represents the income level around which the lower middle class starts (lower middle class being defined here as the group that increases in incidence past a threshold survival income level, in this case above appx. 20k), c is the proportion of households at that income level, and a is a constant of proportionality such that $ab^2 + c$ represents the number of households surviving on practically no income.

Similarly, the exponential can be thought of as ae^{-bx}, where a is a constant of proportionality which corresponds to the theoretical probability of an income of 0 under this model (this constant a is naturally not going to be reflected in reality, since the exponential model only starts to reflect the data past 60k, the income threshold of a middling to upper middle class), and b reflects the rate at which the proportion of people at a certain income level decreases per unit income. In fact, this equation can be re-expressed to a logarithm with base-2 and be understood in terms of an income half-life;

i.e. the amount of income that you need to increase your current income by to fall in half the upper percentile you currently fall in.

A chi-squared goodness of fit-test was performed to assess the suitability of this model, revealing on 20 degrees of freedom (comparing across 21 categories, at interval levels of 10k on the histogram's x-axis) a chi-squared value of 3.60. This is an exceptionally small statistic, meaning the model is an excellent fit for the data, and the variation seen is likely to be observed over 99% of the time owing to observational randomness. This is expected, since the best-fit model was done with spreadsheet software that uses calculus to regress to minimising least squares (the mean squared error function is calculated for proposed models, then derivatives used to minimise the same; numerical iterations are performed until further mean square error values are not significantly smaller). Without performing such checks manually, it is possible to demonstrate that this model makes intuitive sense. Eyeing the data, a quadratic graph for the first section would have vertex (22, 10) and intercept (0, 20), so from $a(x - b)^2 + c$, b is 22 and c is 10; simple equation calculations reveal an a-value of 5/242. Since the second half is asymptotic to the positive x-axis, the equation will have the form of ae^{-bx}, using points (80, 19) and (120, 9) as (x, y) pairs through which the equation must pass and solving simultaneous equations for the values of a and b yields an approximate a-value of 80 and a b-value of ln $2^{0.025}$. The graph:



Not bad for a human! The chi-squared statistic reveals that this model is a good explanation for the data about 80% of the time, so while the computer model is certainly the more fine-tuned and optimised version, this manual implementation does show that a quadratic and exponential piecewise is a good way to model this income distribution.

But of course a fundamental problem remains. A proper probability distribution, even in two halves, should add up to an area of 1, which integrating either of these two will not give. Furthermore, the cutoff point is almost arbitrary, being defined by this year's data – a different year would have this local maximum occur elsewhere. My concern then is that these distributions, although they match this particular year's data well, do not reflect or understand deeper underlying trends (a better function, for instance, might predict the local maximum instead of simply cutting off the data into halves at the local mode).

However, it is undoubtable that we do get a significantly better approximation if we split the graph into two separate distributions. The hump at the beginning and the local

maximum in the middle are not reflective of any sort of sinusoidal trend, since those two maxima are the only two, and do not repeat later. So in looking for a different model that better explains this maximum, I noticed that the graph from 25k onwards looks like it can be modelled quite well as a skewed normal distribution, but since the equation for a skewed normal looks like this:

$$f(x) = \frac{2e^{\left(\frac{-\left(\left(\frac{x-\xi}{\omega}\right)-\mu\right)^2}{2\sigma^2}\right)} \cdot \left(1 + \frac{2}{\sqrt{\pi}} \int_0^{\left(\frac{\alpha(x-\xi)}{\omega\sqrt{2}}\right)} e^{-t^2}\, dt\right)}{2\omega\sigma\sqrt{2\pi}}$$

… and would require access to raw census data for evaluation anyway, I instead set about finding models that matched the overall shape of my data for comparison, and came upon this negative binomial distribution as a possible candidate:

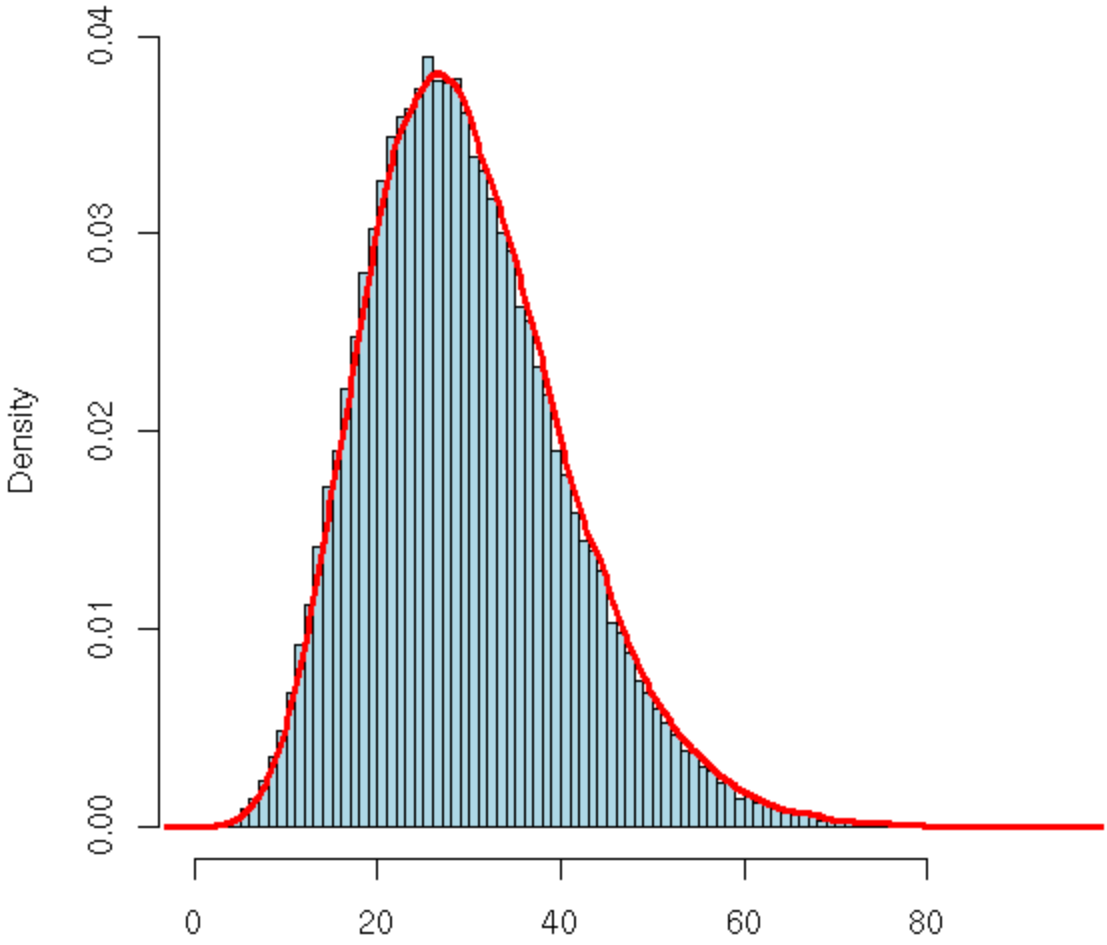


***Fig. 5**: A Negative Binomial Distribution*

A negative binomial distribution tracks the number of trials that can be conducted before a pre-specified number of outcomes is reached.

$$f(k;r,p) \equiv \Pr(X = k) = \binom{k+r-1}{r-1}(1-p)^k p^r$$

To adapt it to this data, we would track how many people must be chosen at random before we find an individual with an income higher than the set threshold (r is the number of successes, k the number of failures, and p the probability of finding a success, where a success is defined as finding an individual that matches our criteria). Running such a simulation is beyond the scope of this project but if conducted would represent the massive disparity inherent in the histogram (e.g. a slight left-shift offers a good match).

Now, the histograms for 1990 and 2019, drawn similarly to the one above for 1967 with income on the horizontal axis and percentage of households within each bracket on the vertical axis:

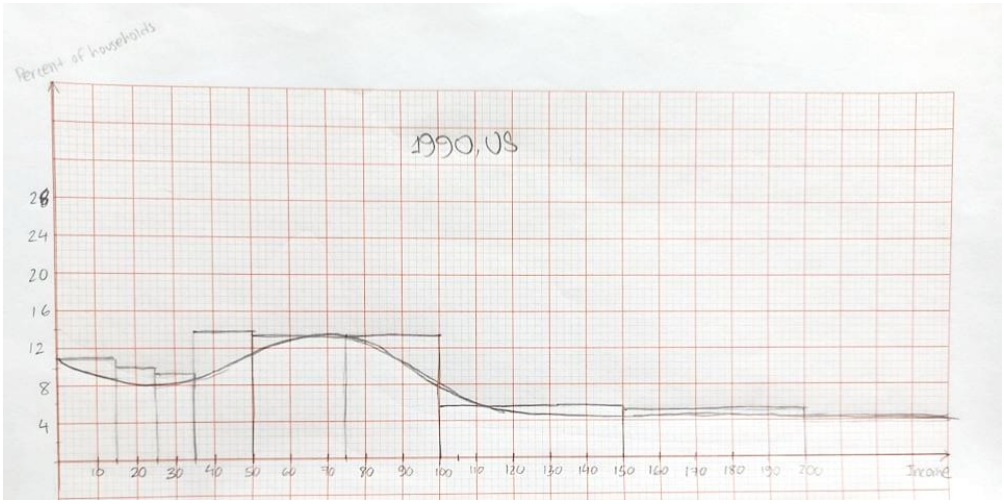


***Fig. 6**: Percent of Households at Specified Income Levels, 1990*

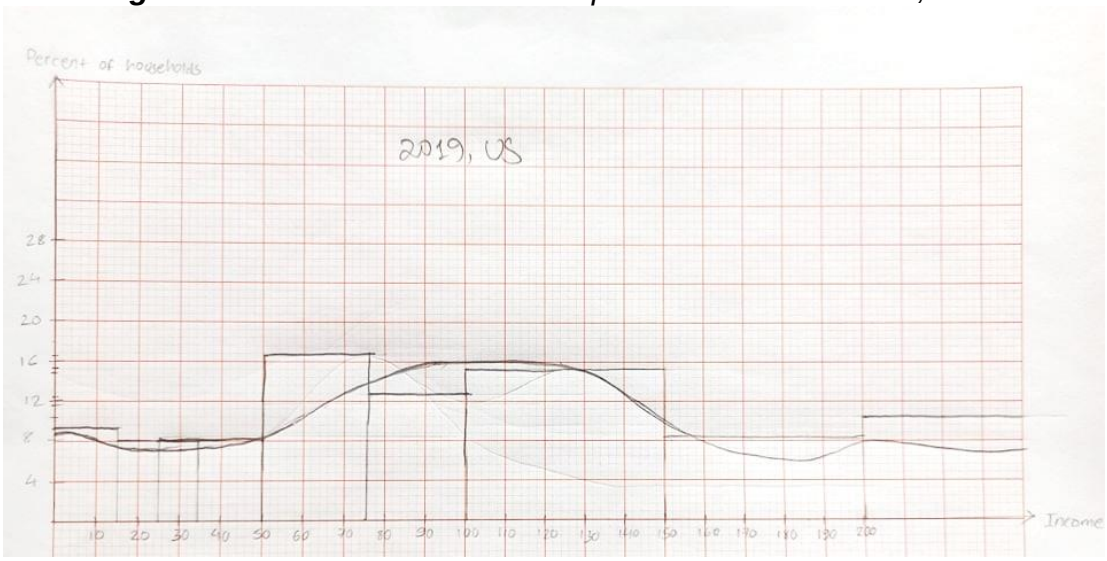


***Fig. 7**: Percent of Households at Specified Income Levels, 2019*

One of the primary differences we see between these distributions and the 1967 one is the flattening of the mode. There are fewer households concentrated in a single bracket, with a much larger middle class and a slightly less pronounced hump of very low-income levels. That said, inequalities of income are still very apparent, with most households not moving past an ability to afford basic necessities and an education – recall that although the median income value is shifting to the right, this accompanies larger increases in prices and CPI and thus actually indicates a lowering of real purchasing power at this same time, as is seen in Fig. 8 below. This is exactly the great lie of averages I wanted to point out with my analysis: that although my income distributions seem like they are going from a negative binomial in 1967 to something that looks more normal or almost uniform around the mode in 2019, in truth, real purchasing power as derived by a comparison with CPI and adjusting for inflation, demonstrated in Figs. 2 and 8, was actually fluctuating and is often decreasing over that time period for most individuals. Despite consistent GDP growth, the group of very low-income households at below 25k a year continued to make up a similar portion of the total distribution across all years – the story of economic growth as given by the simplistic aggregate marker of GDP growth was not their story.

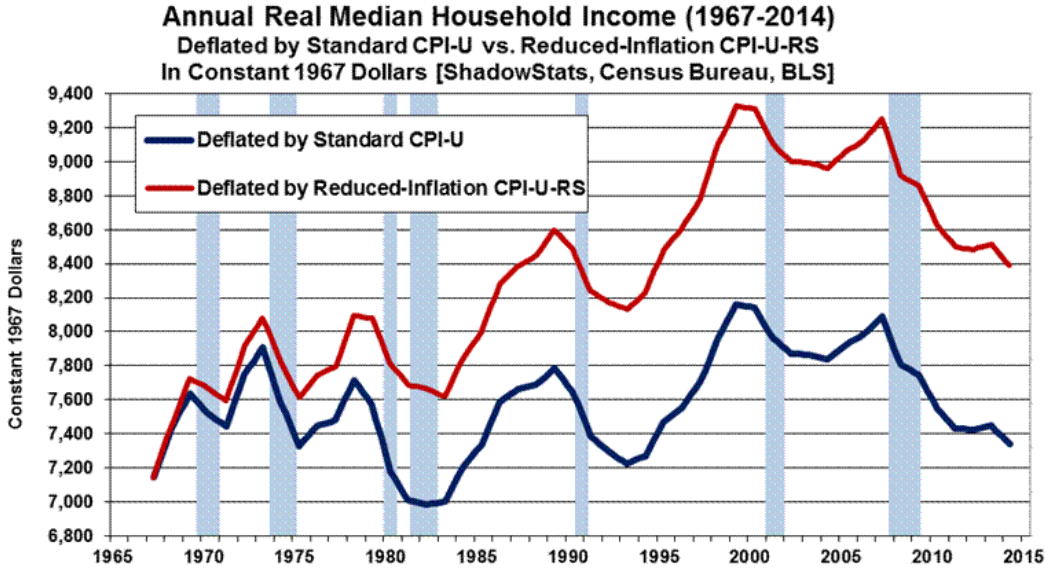


***Fig. 8**: Adjusting Income with CPI, US 1967-2015*
Image courtesy of the US Census Bureau

Thus if and when this project is extended by developing a metric that reflects economic stability and potential for achievement of wants for all individuals, the above analysis indicates that a plot of economic stability vs. frequency would show low stability levels for the vast majority and high stability levels for a very small proportion of people.

Conclusion:

I would like to end this discussion by proposing that we therefore stop speaking of, or at least modify, the current usage of simple indicators like GDP, which can present

misleading and inflated pictures of socio-economic conditions when not used in conjunction with other indices like real purchasing power. Throughout the entirety of the years discussed above, the GDP has almost steadily risen, with slight dips during recessions. This is hilariously bad at demonstrating the real struggles of most people whose work contribute to the GDP and the assets the majority possess, to the extent that if an economics student at the IB level were to say a country is fine economically by coming up with GDP as a metric, he would be failed on his assessment for failing to take stock of important factors that add nuance to the discussion and paint the real picture instead of “it’s great, the country’s been developing steadily”. The country hasn’t been – a tiny portion of it has been, and that’s assuming that all that there is to development is increasing assets.

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Appendix A: Median Incomes of US Households, All Races, 1967 – 2019

Race and Hispanic origin of householder and year	Number (thousands)	Percent distribution							
		Total	Under $15,000	$15,000 to $24,999	$25,000 to $34,999	$35,000 to $49,999	$50,000 to $74,999	$75,000 to $99,999	$100 to $149
ALL RACES									
2019	128,451	100	9.1	8.0	8.3	11.7	16.5	12.3	
2018	128,579	100	10.1	8.8	8.7	12.0	17.0	12.5	
2017 [2]	127,669	100	10.0	9.1	9.2	12.0	16.4	12.4	
2017	127,586	100	10.1	9.1	9.2	11.9	16.3	12.6	
2016	126,224	100	10.4	9.0	9.2	12.3	16.7	12.2	
2015	125,819	100	10.6	10.0	9.6	12.1	16.1	12.4	
2014	124,587	100	11.4	10.5	9.6	12.6	16.4	12.1	
2013 [3]	123,931	100	11.4	10.3	9.5	12.5	16.8	12.0	
2013 [4]	122,952	100	11.3	10.4	9.7	13.1	17.0	12.5	
2012	122,459	100	11.4	10.6	10.1	12.5	17.4	12.0	
2011	121,084	100	11.6	10.2	10.2	13.1	17.2	11.9	
2010 [5]	119,927	100	11.2	10.7	9.4	13.3	16.8	12.4	
2009 [6]	117,538	100	10.4	10.0	9.7	13.2	17.4	12.4	
2008	117,181	100	10.4	10.0	9.4	13.4	17.0	12.5	
2007	116,783	100	10.0	10.0	9.0	12.8	17.3	12.6	
2006	116,011	100	9.6	9.5	9.4	13.0	17.5	12.7	
2005	114,384	100	10.1	9.8	9.5	12.7	17.6	12.9	
2004 [7]	113,343	100	10.3	9.7	9.8	12.8	17.4	12.6	
2003	112,000	100	10.2	9.7	9.7	12.4	17.4	12.7	
2002	111,278	100	9.9	9.8	9.2	13.3	17.4	12.7	
2001	109,297	100	9.7	9.6	9.1	12.9	17.7	12.8	
2000 [8]	108,209	100	9.3	9.3	8.9	13.1	17.7	13.2	
1999 [9]	106,434	100	9.0	9.7	9.1	13.0	17.5	13.3	
1998	103,874	100	9.8	9.8	9.2	12.9	17.5	13.7	
1997	102,528	100	10.3	10.3	9.4	13.2	18.0	13.3	
1996	101,018	100	10.6	10.5	9.6	13.5	17.9	13.5	
1995 [10]	99,627	100	10.5	10.4	10.2	13.1	18.8	13.3	
1994 [11]	98,990	100	11.3	10.8	10.1	13.5	18.4	12.7	
1993 [12]	97,107	100	11.8	10.6	10.1	13.7	18.4	13.1	
1992 [13]	96,426	100	11.8	10.6	10.1	13.5	18.8	13.5	
1991	95,669	100	11.6	10.3	9.8	14.0	19.1	13.5	
1990	94,312	100	11.1	10.1	9.5	13.9	19.5	13.7	

1989	93,347	100	10.8	10.0	9.6	13.4	19.2	13.9	
1988	92,830	100	11.4	9.7	9.9	13.1	19.4	14.0	
1987 [14]	91,124	100	11.6	10.0	9.9	13.6	18.9	14.2	
1986	89,479	100	12.0	10.0	9.9	13.6	19.4	13.9	
1985 [15]	88,458	100	12.1	10.6	10.1	14.3	19.8	13.6	
1984 [16]	86,789	100	12.2	10.8	10.1	14.7	19.7	13.7	
1983	85,407	100	12.7	10.9	10.6	14.8	20.1	13.3	
1982	83,918	100	12.7	10.8	10.9	14.5	20.8	13.4	
1981	83,527	100	12.5	10.9	11.0	14.3	20.6	14.0	
1980	82,368	100	12.2	10.7	10.3	14.6	21.2	14.0	
1979 [17]	80,776	100	11.8	10.2	10.2	14.4	20.5	15.1	
1978	77,330	100	11.6	10.5	10.1	14.2	21.0	14.9	
1977	76,030	100	11.9	11.2	10.5	14.2	21.3	14.7	
1976 [18]	74,142	100	12.0	11.3	10.3	14.8	21.8	14.5	
1975 [19]	72,867	100	12.4	11.4	10.7	14.7	22.1	14.3	
1974 [19, 20]	71,163	100	11.8	10.7	10.0	15.3	22.0	14.9	
1973	69,859	100	11.6	10.8	9.5	14.3	22.2	14.9	
1972 [21]	68,251	100	12.4	10.3	10.0	14.7	22.4	14.7	
1971 [22]	66,676	100	13.3	10.2	10.6	15.2	23.5	13.9	
1970	64,778	100	13.3	10.1	9.7	15.7	24.1	13.7	
1969	63,401	100	13.2	9.9	9.4	16.0	24.3	14.2	
1968	62,214	100	13.4	10.1	10.4	16.5	24.8	13.7	
1967 [23]	60,813	100	14.8	10.2	10.9	16.8	24.8	11.9	